

THE FREE GUIDE · NO. 01

Decode Any *METAR & TAF*

The scariest string in aviation, read slot by slot in plain English. Once you can read one, you can read them all. Printable kneeboard cheat sheet on the last page.

START HERE

It looks like a code. It's actually a sentence.

Weather is the single most-missed area on the FAA knowledge test, and it's almost never because the weather is hard. It's because the format looks like noise.

A METAR is just an observation of what the weather is doing right now at one airport. A TAF is the forecast for that airport over the next 24 to 30 hours. Both are written in the same shorthand, and both follow the **same fixed order** every single time. That order is the whole secret: once you know which slot comes next, the string stops being a wall of letters and starts reading like a sentence.

This guide walks you through that order, slot by slot, with a real example at every step. Work through it once tonight, then pull the actual METAR for your home field tomorrow and read it out loud. That is the moment it clicks.

HOW TO READ THIS GUIDE

Pages 3 and 4 decode a full METAR from left to right. Page 5 covers the remarks and the trickier groups. Pages 6 and 7 do the same for the TAF and its change groups. Page 8 is a one-page cheat sheet built to print and live on your kneeboard. You don't need to memorise anything; you need to recognise the pattern.

THE GOLDEN RULE OF THE SPREAD

If you take one number from this whole guide, take the gap between **temperature and dew point**. When they sit within about 2 to 3°C of each other, the air is nearly saturated: watch for fog and low cloud, especially around dawn. It is the most useful single number on the report.

THE METAR · LEFT TO RIGHT (1 OF 2)

One real report, every slot

**KSF0 301853Z 28012G20KT 10SM FEW015 SCT200 18/12 A2992 RMK
A02 SLP132**

SAN FRANCISCO INTL, THE 30TH AT 1853 ZULU

1 Station KSF0

The ICAO identifier for the reporting airport. US fields start with K. This is San Francisco International.

2 Date and time 301853Z

The 30th day of the month, at 1853 **Zulu** (UTC). Aviation weather always runs on Zulu time, never local. The Z is your reminder.

3 Wind 28012G20KT

Wind from **280°** true at **12 knots**, gusting **20**. Calm reads 00000KT. Variable direction under 6 kt reads VRB. A shifting wind shows as, for example, 280V350 after the speed.

4 Visibility 10SM

Prevailing visibility in **statute miles**. 10SM means 10 or more. Fractions appear as 1/2SM. This is the headline number for your VFR minimums.

WHY ZULU, ALWAYS

A flight can cross several time zones, and a controller in another region has to read the same report you do. One shared clock removes the ambiguity. Get comfortable converting your local time to Zulu now; you will do it on every briefing for the rest of your flying life.

THE METAR · LEFT TO RIGHT (2 OF 2)**1 Weather and obstructions**

When present, this slot describes what's falling or limiting sight. Intensity first: - light, + heavy, none for moderate. Then types: RA rain, SN snow, BR mist, FG fog, TS thunderstorm, SH showers, FZ freezing. So +TSRA is heavy thunderstorm with rain.

2 Sky condition FEW015 SCT200

Coverage in eighths of the sky, with the base height in **hundreds of feet AGL**. FEW015 is few clouds at 1,500 ft; SCT200 is scattered at 20,000 ft. The ladder: **FEW, SCT, BKN, OVC**. BKN or OVC is a ceiling. CLR or SKC means clear; VV means sky obscured.

3 Temperature and dew point 18/12

In **Celsius**, temperature first. A leading M means minus, so M02/M04 is minus 2 over minus 4. Here the spread is 6°C: not close to saturation. Watch this gap close on any night report.

4 Altimeter A2992

The altimeter setting in **inches of mercury**, decimal implied: 29.92. Set this so your altimeter reads true field elevation. A leading Q (for example Q1013) is the same thing in hectopascals, used outside the US.

READ THE SKY LADDER LIKE A DIMMER

FEW, SCT, BKN, OVC simply turn the sky from clear to covered in four steps. The instant you reach **BKN** (broken) you have a ceiling, and a ceiling is the number that decides whether the field is VFR. Train your eye to jump straight to the first BKN or OVC group.

THE WHOLE REPORT, IN ONE PLAIN SENTENCE

Our example reads: "San Francisco, 30th at 1853 Zulu, wind 280 at 12 gusting 20, 10 miles or more, a few clouds at 1,500 and scattered at 20,000, 18 over 12, altimeter 29.92." That is all a METAR ever is.

THE METAR · REMARKS & THE TRICKY BITS

After the RMK, the extras

Everything after RMK is remarks: useful detail added by the station. You don't need them all, but a few are worth knowing on sight.

GROUP	MEANS
A01 / A02	Automated station, without (AO1) or with (AO2) a precipitation sensor
SLP132	Sea-level pressure 1013.2 hPa (prefix 9 or 10 as needed to read near 1013)
P0008	Precipitation since the last report, here 0.08 inches in the past hour
T01830122	Precise temperature and dew point to a tenth: 18.3 over 12.2°C
PK WND	Peak wind since the last observation
WSHFT	A wind shift, often with the time it occurred

THREE TRAPS THAT CATCH STUDENTS

Heights are AGL, not MSL. Cloud and ceiling bases are feet above the ground at that field. **The time is Zulu, not local.** Convert before you reason about it. **Visibility is statute miles in a US METAR** (the SM is stated), while many countries report it in metres, so 9999 abroad means 10 km or more, not a visibility of nine thousand anything.

CAVOK, A WORD YOU'LL MEET ABROAD

Outside the US you may see CAVOK ("ceiling and visibility OK"): visibility 10 km or more, no cloud below 5,000 ft (or the highest minimum sector altitude) and none cumulonimbus, and no significant weather. US reports don't use it, but it's worth recognising the moment you fly internationally.

THE TAF • THE FORECAST (1 OF 2)

Same shorthand, now looking forward

A TAF reads almost exactly like a METAR. The difference is that it forecasts a window of time, and it uses change groups to say "and then this happens."

TAF KSF0 301730Z 3018/0124 28010KT P6SM SCT020

ISSUED THE 30TH AT 1730Z, VALID 30TH 1800Z THROUGH THE 1ST 2400Z

1 Header **TAF KSF0 301730Z**

The label, the station, and the issue time in Zulu. Sometimes TAF AMD for an amended forecast, or TAF COR for a correction.

2 Valid period **3018/0124**

The forecast runs from the **30th at 1800Z** to the **1st at 2400Z**. The format is dayhour/dayhour. Many TAFs cover 24 hours, larger airports 30.

3 The base forecast **28010KT P6SM SCT020**

Read it exactly like a METAR: wind 280° at 10 kt, visibility more than 6 statute miles (P6SM), scattered cloud at 2,000 ft. This is the prevailing forecast until a change group says otherwise.

P6SM, THE TAF'S "PLENTY"

A TAF caps visibility at P6SM, meaning "more than 6 statute miles." Forecasters don't predict exact visibility above that, because for nearly every operation more than 6 miles is simply good. See P6SM and think "visibility is not the problem in this window."

THE TAF • CHANGE GROUPS (2 OF 2)

The words that say "and then"

Everything that makes a TAF look complicated is just these change groups stacked up. Learn the handful below and you can read any forecast in the country.

GROUP	WHAT IT TELLS YOU
FM	From. A clean, rapid change at a stated time. FM0200 means from 0200Z, everything after this replaces what came before
TEMPO	Temporary. Brief fluctuations expected to last under an hour each and less than half the period. TEMPO 3022/3024 applies in that window
BECMG	Becoming. A gradual transition over the stated hours to a new prevailing condition
PROB30 / PROB40	Probability. A 30 or 40 percent chance of the conditions that follow, usually thunderstorms or low visibility
Intensity	Groups still carry intensity, so TEMPO . . . TSRA is a temporary thunderstorm with rain

FM010200 31015G25KT 4SM -SHRA BKN015 PROB30 0104/0108 2SM
TSRA

FROM THE 1ST 0200Z: WIND 310 AT 15 GUST 25, 4 MILES IN LIGHT RAIN SHOWERS, BROKEN AT 1,500. 30 PERCENT CHANCE 0400 TO 0800Z OF 2 MILES IN A THUNDERSTORM WITH RAIN

READ A TAF FOR THE TREND, NOT THE SNAPSHOT

The question is never just "what is forecast." It's "is this window getting better or worse, and will it still be flyable when I arrive and again when I leave." A BECMG to a lower ceiling, or a PROB of thunderstorms over your return time, is the report telling you to keep a plan B.

KEEP · PRINT THIS PAGE

The kneeboard cheat sheet

METAR / TAF QUICK DECODE

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READ IN THIS ORDER

- 1 Station (ICAO, US starts with K)
- 2 Day + time, always Zulu (Z)
- 3 Wind: dir true / speed / Gust KT
- 4 Visibility, statute miles (SM)
- 5 Weather and obstructions
- 6 Sky: coverage + base (hundreds ft AGL)
- 7 Temp / Dew point, Celsius
- 8 Altimeter A###.## inHg
- 9 RMK: remarks

SKY COVERAGE

- FEW** few
SCT scattered
BKN broken (ceiling)
OVC overcast (ceiling)
VV vertical vis, obscured

WEATHER CODES

- / +** light / heavy
RA SN rain, snow
BR FG mist, fog
TS SH thunderstorm, showers
FZ freezing

TAF CHANGE GROUPS

- FM** from (rapid, clean change)
TEMPO brief, under half the window
BECMG gradual transition
PROB 30 or 40 percent chance
P6SM more than 6 SM

NEVER FORGET

- Z** time is Zulu, not local
AGL cloud heights above ground
2-3° tight T/Dew spread, fog watch

NEXT STEP

Reading the report is half the skill. Knowing what the weather will *do*, and making the go/no-go call, is the other half. That's the Aviation Weather Playbook, and the checkride oral is its own field guide. Both are at groundworkaviation.com.

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